## Decision Tree (I)

優點：可解釋性強、效率高

缺點：過擬合、不穩定性（小變動結果不同）、全域最佳解、資料不平衡導致的偏誤

### 決策樹建構

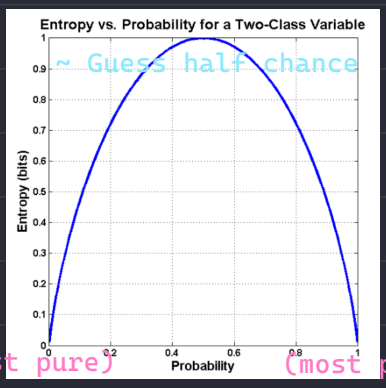
1. 建立一個根節點，並指派所有訓練資料至其中。
2. 根據特定條件，選擇最佳的分枝屬性。　　分枝準則（Splitting criteria）
3. 在根節點為每項分枝值加入分支。
4. 沿著特定分枝路線，將資料分枝至互斥的子集。
5. 為每個葉節點重複步驟2與3，直到達到停止條件為止。　　停止條件 (Stopping) $\text{Gain} = 0$
6. 修剪 (pruning)

### ID3 決策樹演算法 (C4.5)

Entropy (資訊熵)：即不確定性，其值越高則越不確定 e.g. Entropy=1 即半對半錯

$Entropy(S) = -p \cdot \log_2(p) - q \cdot \log_2(q)$ for sample negative p and positive q

Entropy ranges between 0 to 1




~ Guess half chance

(most pure)　　(most pure)

#### Information Gain (訊息增益)

越大即表示資訊越有意義，如果 Entropy = 1 則該節點為葉節點

$$Gain(S, A) = Entropy(S - \sum_{v \in D_A} \frac{|S_v|}{S} Entropy(S_v)$$

# Decision Tree (II)

會考 Gini Index ，因為比 Entropy 的 log 好算

Classification and Regression Trees (CART)

Predictor variables: 因子

Homogeneous：同質性

## Key Ideas

1. Recursive partitioning

   Repeatedly split the records into two parts so as to achieve maximum homogeneit (同質性)

   within the new parts

   (a) 遇到數值，就取相鄰兩兩的平均值切分

   (b) 遇到 enum{A, B, C}，就分成 A & (B || C), B & (A || C), C & (A || B)

   (c) Measuring Impurity - Gini Index (CART)

   $$GI(A) = 1 - \sum_{k=1}^{m} (p_k)^2$$

   $p_k$ : proportion of cases in rectangle A that belong to class k

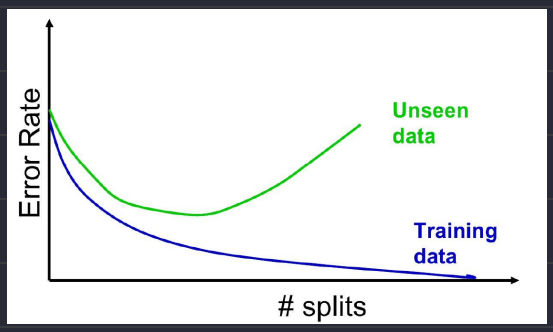
   $GI(A) = 0$ : when all cases belong to same class

   $GI(A) = Max$ : when all class are equally represented (0.5 in binary case)

   (d) Obtain overall impurity measure (weighted avg)

2. Stop condition

   Natural end of process is 100% purity in each leaf




3. Pruning the tree

   Simplify the tree by pruning peripheral branches to avoid overfitting

# Regression Trees

## Construction Steps

1. Choose the split points, usually the average value of each nearby points
2. Choose the split point with minimum Impurity $Impurity = RSS = SSE$

   RSS: Residual Sum of Squares

   SSE: Sum Square of differences bewteen actual val and predict val
3. By doing the step 1 and 2 repeatly, until the Impurity = 0
4. Pruning tree

   If impurity of each leaf node is 0, tree would over fit,

   so we need Tree Score to evaluate the best tree size

   $$\text{Tree Score} = R_\alpha(T) = R(T) + \alpha|\bar{T}|$$

   $R(T)$ → 預測值與實際值的誤差平方 $SSE = \Sigma_{i=0}^{N}(\hat{y} - y)^2$

   $\alpha$ → 懲罰係數，越大剪支後樹的複雜度越低

   $|\bar{T}|$ → 葉節點的數量

   Tree Score 越低越好!
5. How to find the best **$\alpha$**-value?

   a. Construct a max tree $T_{max}$ using both Training data and Testing data

   b. Keep increasing $\alpha$ until pruning leaves give us lower Tree Score.

   → Keep each $\alpha$ value increased says $\alpha_1, \alpha_2, \alpha_3, \alpha_4, ...$

   c. Divide data into testing and training datasets (k-fold cross validation)

   d. Use founded $\alpha$ values to purne each trees (total number: k)

   e. Calclate the average SSR (SSE) value for each forest built with its alpha

   g. Use the alpha value with lowest SSR (SSE) value with the testing data

# Random Forest

## Introduction

The core idea is Bagging (Boostrap + Aggreation)

## Constructing Random Forest

Let L be original learning set composed of p samples

1. Boostrap Sample Set

   Select q samples from L with replacement (May contain repeated samples)

   The typical value of q is p because around 63% of sample would be selected, then rest of others could be used as testing data $L_k = 100(1 - 1/e) \approx 63\%$

2. Bagging

   Feed same input data to k trees (typically 資料量開根號), then vote for result